



Fig. 5: Interrupt routine

that is made using separate IR transmitter and receiver.

Working of the project is simple. Connect the computer to the system using serial port (COM PORT). Open the hyper-terminal and enter the text you want to display (see construction and testing for details). Switch on the motor. You will see the text appear in a short time. The system switches between text and temperature, displaying both for 30 seconds each on the revolving display.

Mechanics

Mechanical arrangement is the most critical part of this project. An AC motor capable of around 1400 revolutions per minute (rpm) is used. The PCB on which the components are soldered is cut into a specific shape in order to improve aerodynamics and reduce any vertical movement. Eight bright LEDs are used, which are mounted on another board that is attached to the main PCB at right angle.

The critical part here is to get the power to the rotating board. There are various ways of doing it. One simple way is to use rechargeable batteries, as shown in the circuit diagram, and



Fig. 6: Carbon brush arrangement

mount them over the board such that they don't come out when the motor rotates. The rugged way of doing this is to use an AC motor with a double shaft. A step-down transformer and a bridge rectifier along with a 7805 IC can be used to get 5 V from the mains supply voltage. Carbon brush, spring and copper wound bush are used to bring the 5V supply on the main revolving PCB as shown in Fig. 6. The motor should be fitted onto a metal frame in order to reduce vibration of the system and get a smooth operation when the motor revolves.

The number of characters that can be displayed depends on radius, i.e., the distance of the LED strip from the centre. It does not depend on the number of LEDs, which is the main advantage of this system. The main thing to note here is that the speed of rotation directly depends on the radius. A bigger radius would mean a higher angular velocity, which may result in more vibrations. Hence care should be taken not to increase the radius length too much.

Software

The software is written in 'C' language and compiled using Keil compiler. Mentioned below are some important functions and their role:

void serial_int (void) interrupt 4 using 0. Interrupt service routine for serial interrupt. It stores indices of the characters entered, by comparing with the ASCII codes array.

void introutine(void) interrupt 0 using 2. Interrupt service routine for external interrupt 0. It is basically a position decoder, which starts timer



interrupt while switching between displaying the message and the temperature.

void timer0(void) interrupt 1 using 3. Interrupt service routine for timer 0 (auto reload mode). It basically displays characters or temperature depending on flag bits, along with blanks in front of the data for scrolling.

Construction and working

An actual-size, single-side PCB for the controller board is shown in Fig. 7 and its component layout in Fig. 8. PCB for the LED strip is shown in Fig. 9 and its component layout in Fig. 10. Suitable connectors are provided on both the PCBs to connect them to each other. To avoid loose connection, it is better that you directly solder the wires at CON2 and CON2 to connect both the PCBs.

A proper weight balance of the main PCB is essential before mounting it on the shaft. After approximately finding the centre such that weights on either side of it are the same, a hole can be drilled in the PCB for fitting it on the shaft. Aerodynamic shape of the PCB also helps to reduce vibrations during the rotation.

There are two critical factors in the hardware assembly of this project: One is the voltage supply that has to be transferred at the top of the board. The other is smooth rotation of the PCB that is mounted on the top of a motor shaft. Any miscalculation of the weight would result in an unbalanced system and the required rpm would not be achieved. Moreover, LEDs have to switch in less than 1/25th of a second,